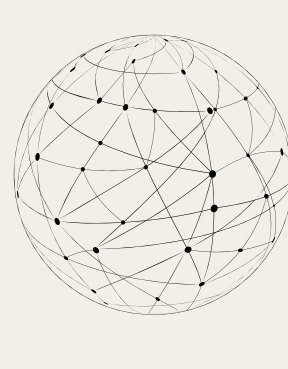


Yolanda Sabuco Garcia

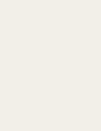
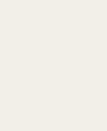
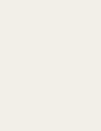


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info@yolandasabuco.io

+41778085336

Luzern, Switzerland



About me

✳️ **Data Strategy & Governance:** proven experience defining data flows within large scale organizations ensuring strict compliance with complex regulations.

✳️ **360° Healthcare Vision:** bridging the gap between technical, medical, and executive departments, leveraging a multidisciplinary background in Medicine, Biomedical Engineering, and Web Engineering.

✳️ **Digitalization of high complexity clinical environments:** specialist in translating complex clinical requirements into digital products across highly specialized medical domains, including Hospital Information Systems, Oncology, Genetics, Dialysis, Diabetes and Optics.

✳️ **Interoperable Systems Architecture:** expertise in designing comprehensive data ecosystems beyond standard implementation, utilizing Enterprise Service Bus, Cloud environments and modern microservices architectures.

✳️ **Healthcare Innovation:** Skilled in effectively integrating AI insights into clinical workflows.

+7 Years experience in Digital Health

5 International companies in Pharma & Healthcare IT

Master's degree in Web Engineering

Polytechnic University of Madrid, Spain

Bachelor's degree in Biomedical Engineering

University of Alicante , Spain

Bachelor's degree in Medicine

Miguel Hernández University, Spain

Studied up to 3rd year

HL7 FHIR® Proficient

Health Level 7

Spanish Security Framework Course (Royal Decree 311/2022, of May 3)

Spanish National Cryptologic Center

Fundamentals SNOMED-CT

SNOMED

Fundamentals of Deep Learning NVIDIA

NVIDIA

Languages

English - Professional

Spanish - Native

Catalan - Native

German - Basic

IT Skills and Projects

Healthcare Interoperability Standards (FHIR, HL7, ISO-11703, DICOM)

Regulations (European Health Data Space, Spanish Security Framework, ISO-27001, GDPR)

Healthcare Interoperability Technologies (FHIR Shorthand language, FHIRPath, HAPI FHIR, Simplifier, Forge, FHIR server for Azure, Lister Hill National Center for Biomedical Communications LHC FHIR Tools)

Healthcare Terminologies, Ontologies and Nomenclators (SNOMED, LOINC, ATC, ICD, Sequence Ontology, HGVS)

Enterprise Service Bus (InterSystems IRIS HealthShare)

Java (Spring Framework)

.NET (Dapper, Entity Framework Core, SQL Server Access)

Unitary test (JUnit, Mockito)

Javascript (Vue.js, Angular, Node.js)

Security (OAuth, OpenID, CAS, JWT, Keycloak)

Artificial Intelligence (Scikit-learn, TensorFlow)

Python (FastAPI, NumPy, Pandas, SciPy, Matplotlib, Seaborn)

Business Intelligence (PowerBI)

Virtualization (Docker, Kubernetes, Oracle VM)

CI/CD (Git, GitHub, GitLab, JFrog, SonarCloud)

Cloud - IaaS and PaaS (FHIR Azure Server, Amazon S3, Heroku)

Relational databases (MariaDB, OracleDB, PostgreSQL)

No-relational databases (MongoDB)

Click for going to the repositories



FHIR® Implementation Guide for the EKM (Elektronische klinische Meldung) of the Swiss Federal Office of Public Health (FOPH).

HL7 FHIR

Clinical Terminologies



FHIR® Roche Diabetes Care Implementation Guide.

HL7 FHIR

Clinical Terminologies



FHIR® Implementation Guide with open source tools

HL7 FHIR

Clinical Terminologies

Docker

CI/CD

Linux



Interoperable platform for monitoring patients with obesity

HL7 FHIR

Clinical Terminologies

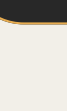
Spring

Angular

ionic

FHIR Azure Server

Heroku



Classifying pathologies using acoustic voice analysis

Artificial Intelligence

Scikit-learn

TensorFlow

Signal Processing

Experience

AHDIS Senior Healthcare Interoperability Specialist

Zurich, Switzerland

Now

Present

PROJECT MANAGEMENT

- **Conceptual work, creating specifications** with FHIR implementation guides, IHE profiles, SwissHDS architecture.
- Consulting related to Digital Health for customers/projects.
- **Technical project management of internal or external Digital Health projects** and support for SwissHDS/EPR integration projects.
- Collaboration on Digital Health seminars, Projectathons, and Connectionthons.

SOFTWARE ENGINEERING

- Software development related to the open source suite from AHDIS and its partners - matchbox, mobile access gateway.

Roche Knowledge Engineer

November 2024 - December 2025

DOMAIN KNOWLEDGE

- **Diabetes.**
- **Remote patient monitoring.**
- **Complex disease management and genetics - cancer.**

PROJECT MANAGEMENT

- **Responsible of the interoperability layer** in the Data Architecture department of Roche Diabetes Care.
- Collaborate with various product teams to shape the development and integration of data models and standardization into the RIS product portfolio and enable FAIR (findable, accessible, interoperable, reusable) data services.
- Collaborate with engineering and business teams and co-drive projects to implement interoperability standards for medical devices and digital products adhering to FAIR principles.
- **Drive the adoption and integration of interoperability standards including terminological resources** (e.g. FHIR, HL7v2, LOINC, SNOMED...) for RIS digital products.
- SAFE development environment, building interoperability solutions while adhering to security protocols and regulatory compliance policies.

SOFTWARE ENGINEERING

- Engage with internal key customers and stakeholders to understand their business needs, pain points using a customer-centric approach.
- **Co-create robust solutions starting out from prototypes and MVPs.**
- **Drive automation of data and software services to achieve continuous delivery** e.g. by setting up code repositories, automation pipelines, quality assurance, etc.

Fresenius Medical Care Data and Connectivity Manager and Security Responsible

March 2024 - November 2024

DOMAIN KNOWLEDGE

- **Nephrology - chronic kidney patient management.** Covering more that 45 dialysis clinics in Spain.

PROJECT MANAGEMENT

- **Design and manage national interoperability strategy with referrals hospitals.**
- Identify potential interconnections between referral hospitals and clinics to define and involve stakeholders in specific new project.
- Evaluate and set up requirements with stakeholders to promote interoperability as part of future tenders.
- **Pre-sales actions and consultancy actions with stakeholders in hospital environment.**
- **Responsible for managing the execution and delivery of projects within budget and scope.**
- **Responsible for managing vendors and external contracts throughout the project lifecycle.**
- **Coordination and supervision the delivery of third-party services to ensure successful project execution.**
- Lead successive phases in a specific project identifying internal people involved and roles.
- **Security Responsible for Fresenius Medical Care Spain.**
- Design and implementation of a Security Management System in Fresenius Medical Care Spain.
- Coordination and facilitation an external audit for Spanish Cybersecurity Framework (Royal Decree 311/2022, of 3 May, National Security Framework) certification, which resulted in a successful certification on the first attempt.
- Preparation of internal documentation (policies and SOPs) and processes, ensuring compliance with the standards required by the auditing body.
- Coordination of global departments and national teams in the company for the achievement of the security certification.

Dedalus Healthcare Interoperability Specialist

October 2022 - March 2024

DOMAIN KNOWLEDGE

- **Electronic Health Record - HCIS.** Solution with a significant impact on digital health, covering over 130 care centers in 3 countries and managing more than 5 million clinical records.

PROJECT MANAGEMENT

- **Definition and implementation of the HCIS Electronic Health Record and compatible products interoperability strategy.**
- **Coordination of architecture team and development teams to implement the interoperability strategy.**

BUSINESS ANALYSIS

- Analysis and documentation of client requirements.
- Design of interoperable solutions based on client requirements.
- **Coordination with clients and stakeholders for implementation.**
- **Coding clinical concepts using medical terminologies and maintaining the clinical concept catalog.**

SOFTWARE ENGINEERING

- Design and development of FHIR Implementation Guide for HCIS Electronic Health Record.
- Design and development and third level support of FHIR and HL7 V2 integrations in Enterprise Service Bus (HealthShare IRIS).

Fujitsu Healthcare Interoperability Specialist

June 2022 - October 2022

DOMAIN KNOWLEDGE

- **Laboratory Information System.**
- **Genetics - Next Generation Sequencing (NGS).**

BUSINESS ANALYSIS

- Analysis and documentation of client requirements.
- Design of interoperable solutions based on client requirements.
- **Coordination with clients and stakeholders for implementation.**
- **Coding clinical concepts using medical terminologies and maintaining the clinical concept catalog.**

SOFTWARE ENGINEERING

- Design and development of FHIR Implementation Guide for exchange information with Laboratory Information System and Hospital Information System.
- Design and development of FHIR and HL7 V2 integrations in Java (Spring Framework).

Atos Healthcare Interoperability Specialist

October 2021 - June 2022

DOMAIN KNOWLEDGE

- **Complex disease management -cancer.**
- **Complex disease management - chronic obstructive pulmonary disease.**

BUSINESS ANALYSIS

- Analysis and documentation of client requirements.
- Design of interoperable solutions based on client requirements.
- **Coordination with clients and stakeholders for implementation.**
- **Coding clinical concepts using medical terminologies and maintaining the clinical concept catalog.**
- **Preparation of technical reports for European Commission.**

SOFTWARE ENGINEERING

- Design and development of FHIR Implementation Guide for exchange information with Electronic Health Records and Artificial Intelligence Modules.
- Design and development of FHIR and HL7 V2 integrations in Java (Spring Framework).

Grupo Miranza Technical Lead VEMoS project

March 2020 - October 2021

DOMAIN KNOWLEDGE

- **Ophthalmology - refractive surgery.**

PROJECT MANAGEMENT

- **Technical Leadership** in VEMoS European Project.
- **Responsible for managing the execution and delivery of projects within budget and scope.**
- **Responsible for managing vendors and external contracts throughout the project lifecycle.**
- **Coordination and supervision the delivery of third-party services to ensure successful project execution.**
- **Preparation of reports for European Commission.**

BUSINESS ANALYSIS

- Analysis and documentation of project requirements.
- **Coordination with clients and stakeholders for implementation.**

SOFTWARE ENGINEERING

- Design and development of VEMoS web application: clinical trial platform and refractive surgery simulation prototype in Javascript (Vue.js) and Java (Spring Framework).
- **Signal Analysis:** Expert in Aberrometry and spectral analysis based on the Fourier Transform.
- **Biophysical Modeling:** Mathematical modeling of biological surfaces and ray-tracing Simulations.

Indra Minsait Full-stack Software Developer

March 2019 - December 2019

DOMAIN KNOWLEDGE

- **Pharmacy Information System.**

SOFTWARE ENGINEERING

- Back-end developer in Java (Spring Framework).
- Front-end developer in Javascript (Sencha Ext.js Framework).

Alicante University Full-stack Software Developer

March 2019 - June 2019

DOMAIN KNOWLEDGE

- **Complex disease management - obesity.**

SOFTWARE ENGINEERING

- Back-end developer in Javascript (Node.js).
- Front-end developer in Javascript, HTML5 and CSS.

Science communications

Click here for reading the communications

[Business Process Management for optimizing clinical processes. A systematic literature review.](#)

De Ramón Alberto, Ruiz-Fernández, Daniel, Sabuco Garcia Yolanda. Health Informatics Journal.

[Classifying pathologies using acoustic voice analysis.](#) Annual Congress of the Spanish Society of Biomedical Engineering.

FCF8 contribution in the [specification and differentiation of the hypothalamus.](#) 16th National Congress of the Spanish Society of Neuroscience 2015.

[Binary classification by voice using machine learning in the Saarbrücken Voice Database.](#) TRANSFORM4EUROPE Doctoral Symposium, Alicante University 2024

[Interoperable platform for monitoring patients with obesity.](#) Master's Thesis, School of Computer Systems Engineering, Technical University of Madrid 2022

[FHIR Interoperability in Personal Health Devices \(PHD Bluetooth profile, ISO 11073 and FHIR\).](#) Bachelor's Degree Thesis, Polytechnic School of the University of Alicante. 2021

Research & Academic Outreach

Academic and Teaching Experience | Telemedicine Biomedical Engineering

Alicante University (Spain).

Invited by the University of Alicante in 2024 and 2025 to give a series of talks to students of Biomedical Engineering on challenges in implementing interoperable systems.

Research Intern | Experimental Embryology Laboratory

Institute of Neurosciences (CSIC-UMH), Alicante, Spain

Summer 2014 & Summer 2015 (6 months total)